Speciation Time & Classifier Free Guidance

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1 Introduction

I recently learned about Speciation Time t_S from a lecture given by Marc Mezard at the Beg Rohu Summer School. To me this gives a better mathematical basis for some work done by Matthieu Wyart "A phase transition in diffusion models reveals the hierarchical nature of data". Basically he shows if you add increasing strength of noise to an image, then run it through a diffusion model, it'll affect small features of the image, and then eventually change the whole class.

So for example, you start with a image of a cat, add some noise then de-noise, it becomes a leopard, do that again it becomes a bear, then a moth.

Wyart does this empirically (although he does give a theoretical model which connects replicates this transition, it seems a little cooked up to me, like where is the score?). I think it'd be interesting to see if we can derive this phenomena completely...

I think a paper which answers this question is

- Biroli et al. "Dynamical Regimes of Diffusion Models". https://arxiv.org/abs/2402.18491.

They derive a pretty general formula for finding these transitions (when does a sample change from one class to another).

However, once you characterize that this transition occurs, you can ask design/engineering questions. The authors use this to cook up a way to explain how/why regular classifier free guidance (CFG) fails at recovering the conditional distribution, and how we can design new CFGs which correct for this.

- Lehman Pavasovic et al. "Classifier-Free Guidance: From High-Dimensional Analysis to Generalized Guidance Forms". https://arxiv.org/abs/2502.07849.

My presentation will go over LP's paper, but bring in some comments from Biroli's paper.

2 Speciation Time

There are three dynamical regimes in the time-reversed process (in the liit of large number of data & large dimension)

1. *Regime I:* At the beginning of the backward process, the random dynamical trajectories have not committed to a particular class of data.
2. *Regime II:* Trajectories have committed to a particular class, and will stay until the ned of the backwards process.
3. *Regime III:* There is no generalization, but instead diffusion models display memorization of the training set.

2.1 Derivation of Speciation Time

Let $\mathcal{D} = \{\vec{a}^\mu\}_\mu$ denote the set of training samples, which we assume were sampled iid from $p_0(\vec{a})$. We can noise our samples using an OU (variance-preserving) process,

$$x_t = \sqrt{e^{-2t}}a + \sqrt{1-e^{-2t}}\eta, \quad \eta \sim \mathcal{N}(0, \mathbb{I}) \tag{2.1}$$

In my notation, the forward process $t \to \infty$ is white noise (base distribution), and the reverse process $t \to 0$ is the data (target distribution). Also for future reference, I'll notate the noise variance as $\Delta_t = 1 - e^{-2t}$.

Let's talk about the probability distributions $p(x_t|a)$. Since this is just two random variables being added, the probability distribution gets convolved:

$$p_t(\vec{x}) = \int d\vec{a}\ p(\vec{x}|\vec{a})p(\vec{a}) \tag{2.2}$$

$$= \int d\vec{a}\ p_0(\vec{a}) \frac{1}{\sqrt{2\pi\Delta_t}^d} \exp\left(-\frac{1}{2}\frac{(\vec{x}-\vec{a}e^{-t})^2}{\Delta_t}\right) \tag{2.3}$$

where

$$p_t(\vec{x}) = \frac{1}{\sqrt{2\pi\Delta_t}^d} \exp\left(-\frac{1}{2}\frac{\vec{x}^2}{\Delta_t} + g(\vec{x})\right) \tag{2.4}$$

where

$$g(\vec{x}) = \log \int d\vec{a}\ p_0(\vec{a}) \exp\left(-\frac{1}{2}\frac{\vec{a}^2 e^{-2t}}{\Delta_t}\right) \exp\left(\frac{e^{-t}}{\Delta_t}\vec{x}\cdot\vec{a}\right) \tag{2.5}$$

We will now work in the large-time limit $t \gg 1$ (near base distribution).

Let $\langle \cdot \rangle$ denote expectation value w.r.t. the effective distribution $p_0(\vec{a}) \exp(-\frac{\vec{a}^2 e^{-2t}}{2\Delta_t})$, which in the large time limit is effectively $p_0(\vec{a})$.

We'll start by Taylor expanding the 2nd exponential $\exp(\frac{e^{-t}}{\Delta_t}\vec{x} \cdot \vec{a}) \approx 1 + \frac{e^{-t}}{\Delta_t}\vec{x} \cdot \vec{a} + ...$

$$g(\vec{x}) = \log\left(1 + \frac{e^{-t}}{\Delta_t}\vec{x} \cdot \langle \vec{a} \rangle + \frac{1}{2}\frac{e^{-2t}}{\Delta_t^2}\langle (\vec{x} \cdot \vec{a})^2 \rangle + \mathcal{O}(xe^{-t})^3\right) \tag{2.6}$$

And now Taylor expand the $\log(1+x) \approx x - x^2/2 + \mathcal{O}(x^3)$

$$g(\vec{x}) = \frac{e^{-t}}{\Delta_t}\vec{x} \cdot \langle \vec{a} \rangle + \frac{1}{2}\frac{e^{-2t}}{\Delta_t^2}\langle (\vec{x} \cdot \vec{a})^2 \rangle - \frac{1}{2}\left(\frac{e^{-t}}{\Delta_t}\vec{x} \cdot \langle \vec{a} \rangle + \frac{1}{2}\frac{e^{-2t}}{\Delta_t^2}\langle (\vec{x} \cdot \vec{a})^2 \rangle\right)^2 + \mathcal{O}(xe^{-t})^3 \tag{2.7}$$

$$= \frac{e^{-t}}{\Delta_t}\vec{x} \cdot \langle \vec{a} \rangle + \frac{1}{2}\frac{e^{-2t}}{\Delta_t^2}\left[\langle (\vec{x} \cdot \vec{a})^2 \rangle - (\vec{x} \cdot \langle \vec{a} \rangle)^2\right] + \mathcal{O}(xe^{-t})^3 \tag{2.8}$$

$$= \frac{e^{-t}}{\Delta_t}\sum_{i=1}^{d} x_i \langle a_i \rangle + \frac{1}{2}\frac{e^{-2t}}{\Delta_t}\sum_{ij=1}^{d} x_i x_j [\langle a_i a_j \rangle - \langle a_i \rangle \langle a_j \rangle] + \mathcal{O}(xe^{-t})^3 \tag{2.9}$$

Plugging this approximation into $p_t(\vec{x})$

$$\log p_t(\vec{x}) = C + \frac{e^{-t}}{\Delta_t}\sum_i x_i \langle a_i \rangle - \frac{1}{2\Delta_t}\sum_{ij} x_i M_{ij} x_j + \mathcal{O}(xe^{-t})^3 \tag{2.10}$$

$$\text{s.t. } M_{ij} = \delta_{ij} - e^{-2t}[\langle a_i a_j \rangle - \langle a_i \rangle \langle a_j \rangle] \tag{2.11}$$

Interpreting the matrix M, a couple things to note

1. That second term, that's just the empirical covariance matrix of the data! That is $\langle a_i a_j \rangle - \langle a_i \rangle \langle a_j \rangle = \Sigma_{ij}$
2. This is actually just the Hessian of $\log p_t(\vec{x})$ (in the limiting case $t \gg 1$, near target distribution).
3. If the Hessian of $\log p_t(\vec{x})$ is positive-semi-definite, then the distribution is log concave. Imagine the target distribution is multimodal (i.e. multiple classes, dog image vs cat image), so while it's still log-concave, we can think of this as the samples not committing to a particular mode.

When $\log p_t(x)$ is no longer convex, meaning samples have committed to a path, I'd like to give that a special name. We'll denote this time as t_S the **speciation time**.

We can derive this by figuring out when the Hessian M is no longer positive-semi-definite. You can do this via operator norm, this gets ya $e^{-2t_s} \leq ||\Sigma||_2^{-1}$. Writing it out in a nice way:

$$\boxed{t_s = \frac{1}{2}\log \lambda_{max}(\Sigma)} \tag{2.12}$$

3 Classifier Free Guidance

Consider generating samples using Langevin dynamics

$$dx_\tau = x_\tau d\tau + 2s_\tau(x) + \sqrt{2}dB_\tau \tag{3.1}$$

where $\tau = t_f - t$ denotes the reverse time, and $s_\tau(x) = \nabla_x \log p_t(x)$ is the score.

Classifier Free Guidance does not do that, rather it directs the diffusion in a manner proportional to the difference between the conditional and unconditional score.

$$s_t^{CFG}(x|c) = s_t(x|c) + \omega[s_t(x|c) - s_t(x)] \tag{3.2}$$

$$\nabla_x(\log p_t(x|c) - \omega[\log p_t(x|c) - \log p_t(x)]) \tag{3.3}$$

So to generate samples, you get your lagenvin dynamics equation (3.1) and swap out $s_\tau(x)$ for $s_t^{CFG}(x|c)$.

CFG offers practical advantages (i.e. works on unlabeled data), however a key question remains: "Is CFG is capable of accurate generating samples from the target distribution?"

3.1 Worked Example: Symmetric Gaussian Mixtures

3.1.1 Setup

Consider a mixture of two d-dimensional gaussians centered at $\pm\vec{\mu}$

$$p_0(\vec{x}) = \frac{1}{2(\sqrt{2\pi\sigma^2})^d}\left(e^{-(\vec{x}-\vec{\mu})^2/2\sigma^2} + e^{-(\vec{x}+\vec{\mu})^2/2\sigma^2}\right) \tag{3.4}$$

Noising this with an OU process, we find the probability $p_t(\vec{x})$

$$p_t(\vec{x}) = \frac{1}{2(\sqrt{2\pi\Gamma_t})^d}\left(e^{-(\vec{x}-\vec{\mu}e^{-t})^2/2\Gamma_t^2} + e^{-(\vec{x}+\vec{\mu}e^{-t})^2/2\Gamma_t^2}\right) \tag{3.5}$$

$$\log p_t(\vec{x}) = -\frac{\vec{x}^2}{2\Gamma_t} + \log\cosh\left(\vec{x}\cdot\vec{\mu}\frac{e^{-t}}{\Gamma_t}\right) + \text{Constant w.r.t. } \vec{x} \tag{3.6}$$

$$\nabla_x \log p_t(\vec{x}) = -\frac{\vec{x}}{\Gamma_t} + \vec{\mu}\frac{e^{-t}}{\Gamma_t}\tanh\left(\vec{x}\cdot\vec{\mu}\frac{e^{-t}}{\Gamma_t}\right) \tag{3.7}$$

where $\Gamma_t = \sigma^2 e^{-2t} + \Delta_t$ We can also ask, what's the conditional distribution's score

$$\nabla_x \log p_t(\vec{x}|\pm\vec{\mu}) = \frac{-\vec{x} \pm \vec{\mu}e^{-t}}{\Gamma_t} \tag{3.8}$$

Plugging this into the CFG score, we get

$$s_t^{CFG}(x|c=\pm\mu) = \frac{-x \pm \mu e^{-t}}{\Gamma_t} + \omega[\frac{-x \pm \mu e^{-t}}{\Gamma_t} + \frac{x}{\Gamma_t} + \mu\frac{e^{-t}}{\Gamma_t}\tanh(x^T\mu\frac{e^{-t}}{\Gamma_t})] \tag{3.9}$$

Rearranging, we can see that the residual that CFG has from the true conditional score

$$s_t^{CFG}(x|c=\pm\mu) - s_t(x|c=\pm\mu) = \underbrace{\mu\ \omega\frac{e^{-t}}{\Gamma_t}[\pm 1 + \tanh(x^T\mu\frac{e^{-t}}{\Gamma_t})]}_{\text{score gap}} \tag{3.10}$$

3.1.2 Speciation Time

We can now ask our self, what is the role of CFG at different times? The only natural time-scale we have is the speciation time, so we can ask, what happens before, at, and after?

To get there, we need to compute the speciation time.

$$t_S = \frac{1}{2} \log \lambda_{max}(\Sigma) \tag{3.11}$$

The covariance has a simple expression

$$\Sigma = \mathbb{E}[XX^T] - \underbrace{\mathbb{E}[X]\mathbb{E}[X^T]}_{\text{0, by symmetry}} \tag{3.12}$$

$$= \Sigma + \mu\mu^T \tag{3.13}$$

$$= \sigma^2 \mathbb{I} + \mu\mu^T \qquad \text{We choose unit covariance WLOG} \tag{3.14}$$

The highest eigenvalue is $\lambda_{max}(\sigma) = \sigma^2 + \|\mu\|^2 := \sigma^2 + d\ \tilde{\mu}^2$. You should interpret $\tilde{\mu}$ as the per-coordinate RMS $\tilde{\mu} \equiv \|m\|/\sqrt{d}$. They choose this because you assume the entires of μ are iid with variance order 1.

> You can interpret this as you're getting more information as your data distribution gains more dimension (i.e. images w/ more resolution). You can choose the scaling $\tilde{\mu} \sim \mathcal{O}(1/\sqrt{d})$ if you want to make statements that you don't learn more information as you gain more dimensions...
>
> Their asymptotic results will vary depending on this parameterization...

Therefore the speciation time for this problem is

$$t_S = \frac{1}{2} \log d + \frac{1}{2} \log \left(\mu^2 + \frac{\sigma^2}{d} \right) \tag{3.15}$$

In the paper, they analyze this in the $d \gg 1$ limit, yielding $t_S = \frac{1}{2} \log d$.

3.1.3 Discussion

Generically, we'll do all this analysis in the direction $q := x \cdot \hat{\mu}$. This is because the trasverse direction (directions orthogonal to μ) don't have interesting dynamics.

Before Speciation Time t_S

> Let's analyze what CFG does prior to speciation time (meaning $t \gg 1$). In this limit $\Gamma_t = 1 - e^{-2t} \approx 1$. The only interesting dynamics occurs in the $\hat{m} = \vec{\mu}/\|\mu\|$ direction.
>
> $$\vec{s}_t^{CFG} \cdot \hat{m} \tag{3.16}$$
>
> Now let's plot the negative log probability in this direction $\vec{x} \cdot \hat{m} := q$
>
> $$-\log p_\tau^{CFG}(q|c = \pm\mu) = \underbrace{(\frac{1}{2}q^2 \mp q\|\mu\|e^{-\tau})}_{\text{Classifier's Log Prob}} + \omega\|\mu\| \underbrace{\left[\mp q e^{-\tau} + \log\cosh(q\|\mu\|e^{-\tau})\right]}_{\text{Extral Log Prob due to CFG}} + \text{Constant} \tag{3.17}$$
>
> See an interactive plot of this at https://www.desmos.com/calculator/vao4sxx6ss.

Notice, when $\omega \neq 0$, you don't recover the correct mean.

Takeaway: **Prior to speciation time, CFG aids in speeding up convergences to the target class.**

Right before speciation time:
The takeaway: **Right before speciation time, the guided path $s^{CFG}(x|c)$ realigns with the unguided path $s(x|c)$ which generates the correct conditional distribution.**

After Speciation Time
After speciation time, samples have committed to a class / path. Hence they are aligned with the mode $x \cdot m \sim \mathcal{O}(d)$, therefore $c - \tanh\left(x \cdot m \frac{e^{-t}}{\Gamma_t}\right) \approx 0$. Therefore the score gap $\delta s_t \approx 0$, meaning the CFG score well approximates the correct conditional score $s^{CFG}(x|c) \approx s(x|c)$.

The take away: **after speciation time, extra corrections from the guidance turn off**.

3.2 Generic Claims

1. **When is guidance helpful:** In the two gaussian mixture, the score gap $\delta s_t(x) = \|\nabla_x \log p_t(x|c) - \nabla_x \log p_t(x)\|$ is tiny at the start (both look like gaussian noise), then increases during regime I (prior to speciation, class undecided), and then collapses back to zero after speciation (class decided). So, **guidance should be on in regime I, and then switch off near/after the speciation time t_S**.

2. Bias of traditional CFG: The residual of the CFG score and true conditional score is linear w.r.t. δs_t

$$s_t^{CFG}(x) = s_t(x|c) + \omega[s_t(x|c) - s_t(x)] \tag{3.18}$$
$$s_t^{CFG}(x) - s_t(x|c) = \omega[s_t(x|c) - s_t(x)] \tag{3.19}$$
$$\|s_t^{CFG}(x) - s_t(x|c)\| = |\omega|\delta s_t \tag{3.20}$$

   In our treatment of the gaussian mixture (in finite dimension), we saw $\delta s_t \sim \mathcal{O}(d^{-1/2})$ (didn't vanish quick enough). So we want a scheme which (exponentially) suppresses δs_t. The take away is to **not keep a linear push when δs_t is small**.

3. Combining (1, guidance help in regime I & switch off) and (2, dampen when guidance shouldn't act)

$$S_t^{NL-CFG}(x) - s_t(x|c) = [s_t(x|c) - s_t(x)]\phi_t(\delta s_t) \tag{3.21}$$

   where $\lim_{x\to 0} x\phi_t(x) = 0$ (ensures (2)). To ensure (1) we just have to cook up the correct ϕ_t which has an amplifying affect.

The authors propose using a power-law guidance $\phi_t(x) = \omega x^\alpha$ (where $\alpha > -1$, to satisfy the limit condition).

4 Results

4.1 Numerics of Gaussian Mixture Model

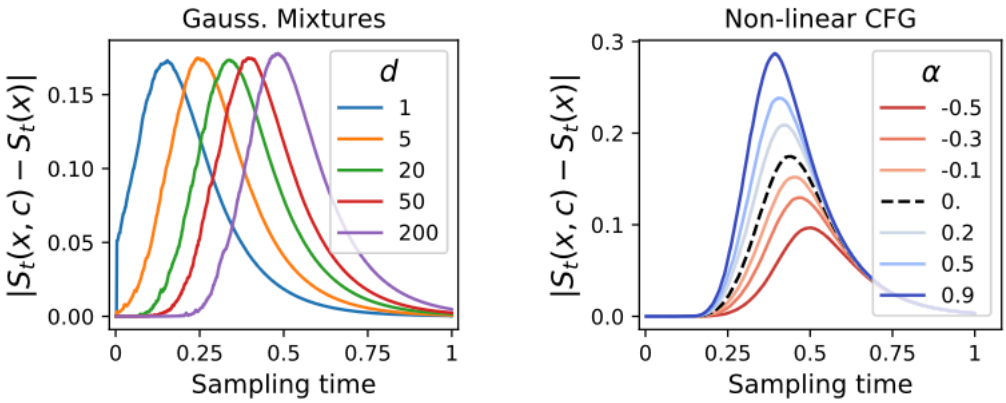
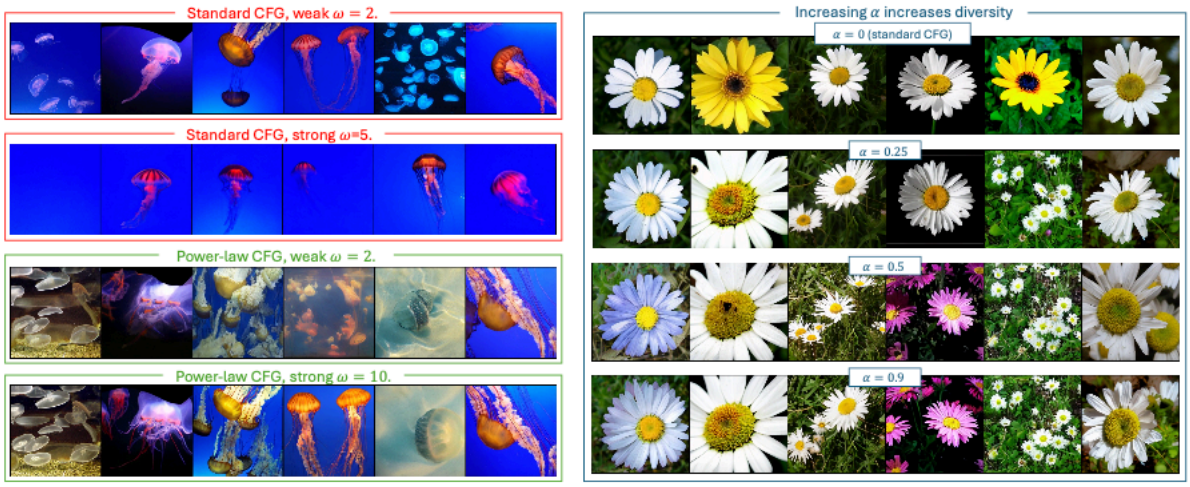


Figure 1: This is Figure 4 in Kruno's paper. They do the symmetric gaussian mixture model that I talked about, and do $s_t^{CFG}(x(t)|c)$ where the trajectory $x(t)$ is generated by Langevin Dynamics $dx_t = (x_t + 2s_t^{CFG}(x_t|c))dt + \sqrt{2}dW_t$. They plot the projection $x \cdot \hat{m}$.

4.2 Images (cause of a diffusion paper isn't complete without images)

See all images on the paper (duh), but here's a screenshot of nice examples.



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